QUBO Formulation: Top Eigencentrality Node Selection Problem

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Problem Statement. Given a graph $G = (V, E)$ with $N = |V|$ nodes, and a pre-computed eigencentrality score $c_i \in \mathbb{R}$ for each node $i \in V$, the objective is to select exactly τ nodes such that the sum of their eigencentrality scores is maximized.

Decision Variables. For each node $i \in V$, introduce a binary decision variable $x_i \in \{0, 1\}$, where $x_i = 1$ indicates that node i is selected, and $x_i = 0$ indicates that it is not selected.

QUBO Derivation. **Step 1.** The original optimization problem is $\max_{\mathbf{x}} \sum_{i \in V} c_i x_i$ subject to $\sum_{i \in V} x_i = \tau$. To cast this as a minimization problem suitable for QUBO, we negate the objective:

$$\min_{\mathbf{x}} \left(- \sum_{i \in V} c_i x_i \right). \tag{1}$$

Step 2. The cardinality constraint is expressed algebraically as:

$$\sum_{i \in V} x_i = \tau. \tag{2}$$

Step 3. We convert the equality constraint into a quadratic penalty term using a Lagrange multiplier $\lambda > 0$:

$$P(\mathbf{x}) = \lambda \left(\sum_{i \in V} x_i - \tau \right)^2. \tag{3}$$

Expanding the square yields:

$$P(\mathbf{x}) = \lambda \left(\sum_{i \in V} x_i^2 + \sum_{i \in V} \sum_{j \in V, j \neq i} x_i x_j - 2\tau \sum_{i \in V} x_i + \tau^2 \right). \tag{4}$$

Applying the idempotent property of binary variables, $x_i^2 = x_i$, the expression simplifies to:

$$P(\mathbf{x}) = \lambda \left((1 - 2\tau) \sum_{i \in V} x_i + \sum_{i \in V} \sum_{j \in V, j \neq i} x_i x_j + \tau^2 \right). \tag{5}$$

Step 4. Combining the negated objective and the penalty term gives the full QUBO Hamiltonian $H(\mathbf{x})$:

$$H(\mathbf{x}) = -\sum_{i \in V} c_i x_i + \lambda \left((1 - 2\tau) \sum_{i \in V} x_i + \sum_{i \in V} \sum_{j \in V, j \neq i} x_i x_j + \tau^2 \right). \tag{6}$$

Step 5. Collecting coefficients for linear and quadratic terms, and expressing the result in standard QUBO form $H(\mathbf{x}) = \sum_{i \leq j} Q_{ij} x_i x_j + c$, we read off the closed-form entries:

$$Q_{i,i} = -c_i + \lambda(1 - 2\tau), \quad \forall i \in V, \tag{7}$$

$$Q_{i,j} = 2\lambda, \quad \forall i, j \in V, i \neq j, \tag{8}$$

$$c = \lambda \tau^2. \tag{9}$$

Penalty Weight Justification. We set the penalty weight to $\lambda = 1 + \max_{i \in V} |c_i|$. The maximum possible improvement in the objective function achievable by flipping any single variable is bounded by $\max_{i \in V} |c_i|$. Violating the cardinality constraint by a single unit changes the penalty term by at least λ. Since $\lambda > \max_{i \in V} |c_i|$, the energy cost of any constraint violation strictly exceeds the maximum possible gain from the objective function. This guarantees that the optimizer will prioritize feasibility over objective improvement.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\{0, 1\}^N$ recovers the optimal solution to the original problem. First, any assignment satisfying $\sum_{i \in V} x_i = \tau$ incurs zero penalty, reducing $H(\mathbf{x})$ to $-\sum_{i \in V} c_i x_i$; thus, the minimum over feasible assignments corresponds exactly to the maximum eigencentrality sum. Second, any infeasible assignment incurs a penalty strictly greater than the maximum possible reduction in the objective term. Consequently, the global minimum of $H(\mathbf{x})$ cannot be infeasible, and must therefore be a feasible assignment that maximizes the original objective.

Illustrative Example. Consider a graph with $N = 3$ nodes, target cardinality $\tau = 2$, and eigencentrality scores $c_0 = 0.5$, $c_1 = 0.8$, $c_2 = 0.3$. The penalty weight is $\lambda = 1 + \max\{0.5, 0.8, 0.3\} = 1.8$. Substituting these values into the general formulas yields the concrete Hamiltonian:

$$H(x_0, x_1, x_2) = -0.5x_0 - 0.8x_1 - 0.3x_2 + 1.8\left((1 - 4)(x_0 + x_1 + x_2) + 2(x_0x_1 + x_0x_2 + x_1x_2) + 4\right) \tag{10}$$

$$= -0.5x_0 - 0.8x_1 - 0.3x_2 - 5.4(x_0 + x_1 + x_2) + 3.6(x_0x_1 + x_0x_2 + x_1x_2) + 7.2 \tag{11}$$

$$= -5.9x_0 - 6.2x_1 - 5.7x_2 + 3.6(x_0x_1 + x_0x_2 + x_1x_2) + 7.2. \tag{12}$$

The corresponding Q matrix entries and offset are $Q_{0,0} = -5.9$, $Q_{1,1} = -6.2$, $Q_{2,2} = -5.7$, $Q_{i,j} = 3.6$ for $i \neq j$, and $c = 7.2$. Enumerating feasible assignments (those with exactly two variables equal to 1), the assignment $\mathbf{x}^* = (0, 1, 1)$ yields the minimum energy $H(\mathbf{x}^*) = -5.9(0) - 6.2(1) - 5.7(1) + 3.6(1) + 7.2 = -0.3$, corresponding to the optimal node selection $\{1, 2\}$ with maximum centrality score $0.8 + 0.3 = 1.1$.